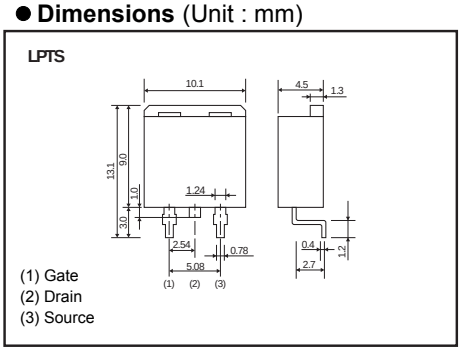
10V Drive Nch MOSFET

R5005CNJ

● Structure
Silicon N-channel MOSFET

● Features
1) Low on-resistance.
2) High-speed switching.
3) Wide range of SOA.
4) Drive circuits can be simple.
5) Parallel use is easy.

● Dimensions (Unit : mm)

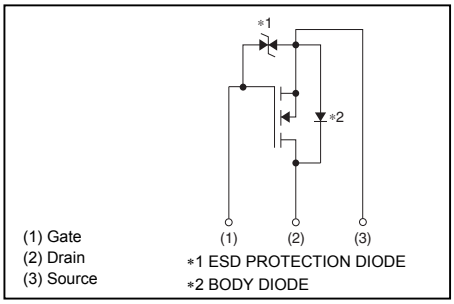


● Application
Switching

● Packaging specifications

Type	Package	Taping
	Code	TL
	Basic ordering unit (pieces)	1000
R5005CNJ		○

● Inner circuit



● Absolute maximum ratings (Ta = 25°C)

Parameter		Symbol	Limits	Unit
Drain-source voltage		V_{DSS}	500	V
Gate-source voltage		V_{GSS}	±30	V
Drain current	Continuous	I_D *3	±5	A
	Pulsed	I_{DP} *1	±20	A
Source current (Body Diode)	Continuous	I_S *3	5	A
	Pulsed	I_{SP} *1	20	A
Avalanche current		I_{AS} *2	2.5	A
Avalanche energy		E_{AS} *2	1.6	mJ
Power dissipation		P_D *4	40	W
Channel temperature		T_{ch}	150	°C
Range of storage temperature		T_{stg}	−55 to +150	°C

*1 Pw≤10μs, Duty cycle≤1%

*2 L ≒ 500μH, V_{DD}=50V, R_G=25Ω, T_{ch}=25°C

*3 Limited only by maximum temperature allowed.

*4 T_C=25°C

● Thermal resistance

Parameter	Symbol	Limits	Unit
Channel to Case	$R_{th\ (ch-c)}$	3.125	°C / W

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● **Electrical characteristics** (Ta = 25°C)

Parameter	Symbol	Min.	Typ.	Max.	Unit	Conditions
Gate-source leakage	I_{GSS}	-	-	±10	μA	V_{GS}=±25V, V_{DS}=0V
Drain-source breakdown voltage	$V_{(BR)DSS}$	500	-	-	V	I_D=1mA, V_{GS}=0V
Zero gate voltage drain current	I_{DSS}	-	-	100	μA	V_{DS}=500V, V_{GS}=0V
Gate threshold voltage	$V_{GS\ (th)}$	2.5	-	4.5	V	V_{DS}=10V, I_D=1mA
Static drain-source on-state resistance	$R_{DS\ (on)}$*	-	1.3	1.6	Ω	I_D=2.5A, V_{GS}=10V
Forward transfer admittance	$\|Y_{fs}\|$*	1.5	2.7	-	S	V_{DS}=10V, I_D=2.5A
Input capacitance	C_{iss}	-	320	-	pF	V_{DS}=25V
Output capacitance	C_{oss}	-	180	-	pF	V_{GS}=0V
Reverse transfer capacitance	C_{rss}	-	15	-	pF	f=1MHz
Turn-on delay time	$t_{d(on)}$ *	-	20	-	ns	V_{DD}≒250V, I_D=2.5A
Rise time	t_r *	-	25	-	ns	V_{GS}=10V
Turn-off delay time	$t_{d(off)}$ *	-	40	-	ns	R_L=100Ω
Fall time	t_f *	-	20	-	ns	R_G=10Ω
Total gate charge	Q_g *	-	10.8	-	nC	V_{DD}≒250V
Gate-source charge	Q_{gs} *	-	3.2	-	nC	I_D=5.0A
Gate-drain charge	Q_{gd} *	-	4.4	-	nC	V_{GS}=10V

*Pulsed

●**Body diode characteristics** (Source-Drain)

Parameter	Symbol	Min.	Typ.	Max.	Unit	Conditions
Forward Voltage	V_{SD} *	-	-	1.5	V	I_S=5.0A, V_{GS}=0V

*Pulsed

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●Electrical characteristic curves

Fig.1 Typical Output Characteristics (Ⅰ)

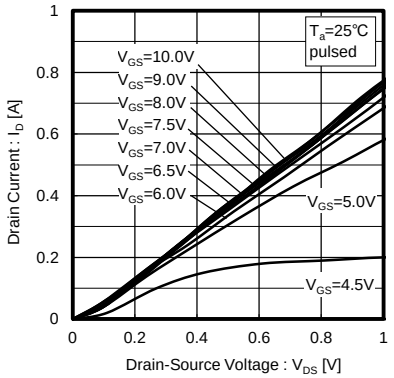


Fig.2 Typical Output Characteristics (Ⅱ)

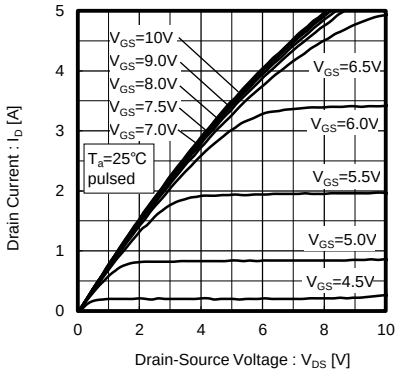


Fig.3 Typical Transfer Characteristics

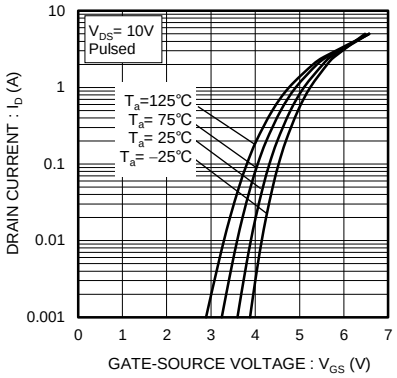


Fig.4 Gate Threshold Voltage vs. Channel Temperature

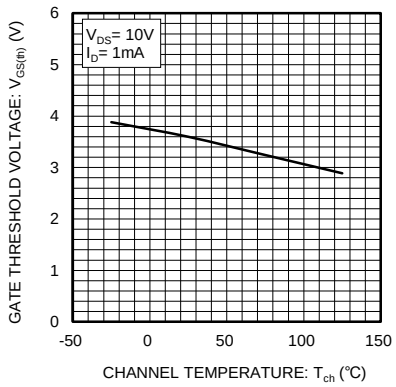


Fig.5 Static Drain-Source On-State Resistance vs. Drain Current

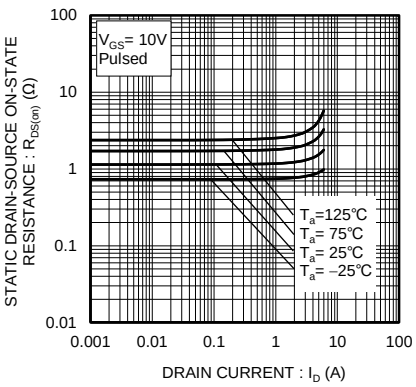


Fig.6 Static Drain-Source On-State Resistance vs. Gate Source Voltage

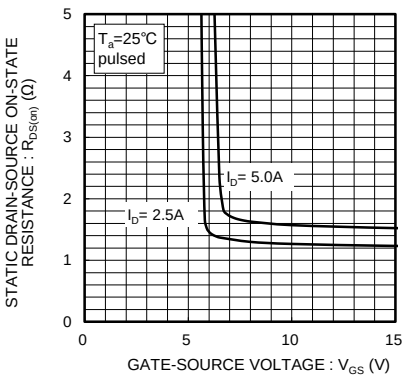


Fig.7 Static Drain-Source On-State Resistance vs. Channel Temperature

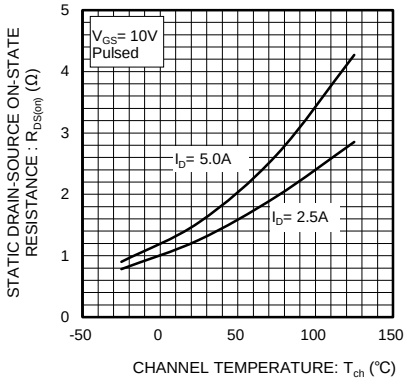


Fig.8 Forward Transfer Admittance vs. Drain Current

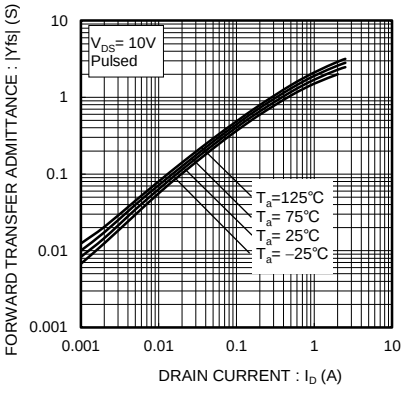
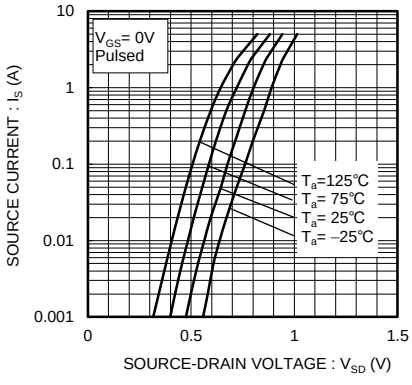


Fig.9 Reverse Drain Current vs. Sourse-Drain Voltage



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Fig.10 Typical Capacitance vs. Drain-Source Voltage

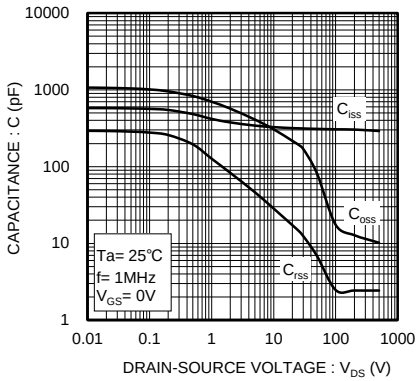


Fig.11 Reverse Recovery Time vs.Source Current

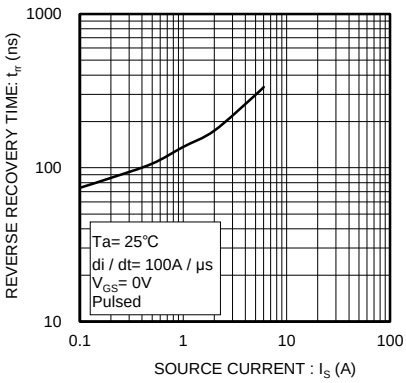


Fig.12 Switching Characteristics

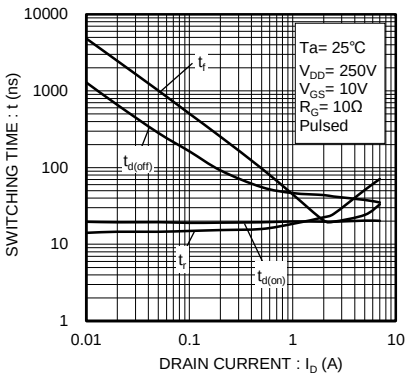
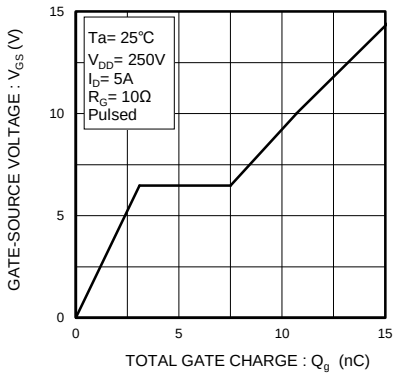


Fig.13 Dynamic Input Characteristics



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● Measurement circuits

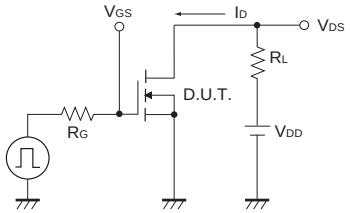


Fig.1-1 Switching Time Measurement Circuit

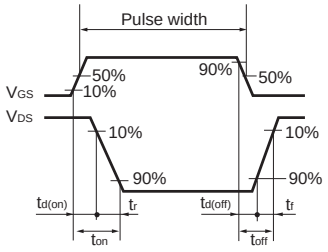


Fig.1-2 Switching Waveforms

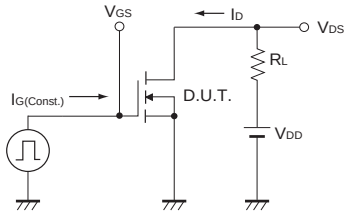


Fig.2-1 Gate Charge Measurement Circuit

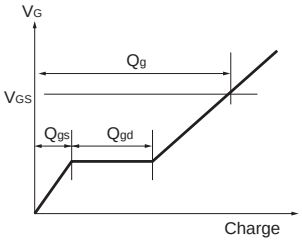


Fig.2-2 Gate Charge Waveform

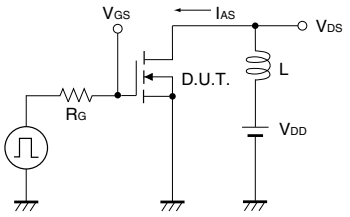


Fig.3-1 Avalanche Measurement Circuit

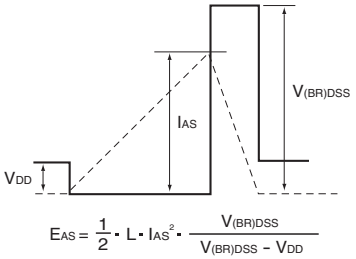


Fig.3-2 Avalanche Waveform

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